

WILHELM

Unified Enterprise AI Agent

KOHLER Hackathon — Track 3

CORE APPROACH

Wilhelm is a multi-turn conversational agent that routes employee queries across enterprise knowledge domains, answers strictly from retrieved source documents rather than general model knowledge, and reformats any answer on demand — JSON, XML, an Excel file, or a ready-to-send email draft — without needing to re-ask the question.

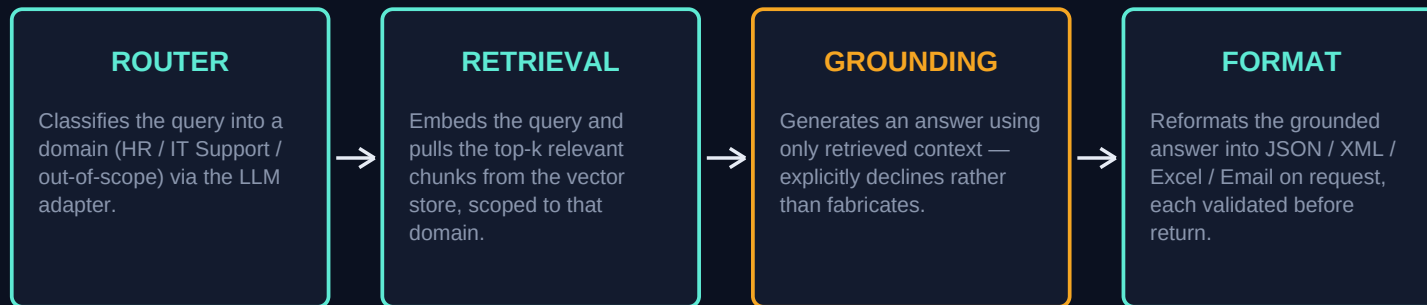
Grounded, not hallucinated

Any LLM · Any vector store

Format-on-demand

System Architecture

Every query flows through a fixed, visible pipeline — shown live in the UI's trace panel.



PROVIDER-AGNOSTIC ADAPTER LAYER

Every stage above calls `generateText()` / `embedText()` / `queryVectorStore()` — never a provider SDK directly.

Swappable via `.env`: OpenAI, Gemini, or any local OpenAI-compatible model (Ollama, LM Studio, vLLM) — plus Chroma or Pinecone.

Session memory persists prior turns, so follow-ups like “now give me that as JSON” resolve without repeating the question.

Tech Stack

Built to swap providers without touching application logic.

FRONTEND

- React + Vite
- Dispatch-console UI (dept rail, trace panel)
- Live-animated pipeline trace on query submit

BACKEND

- Python + FastAPI
- Router / Retrieval / Grounding / Format modules
- Session-scoped conversation memory

LLM LAYER

- Provider-agnostic adapter
- OpenAI · Gemini · local (Ollama / LM Studio / vLLM)
- One function call: generateText()

DATA LAYER

- sentence-transformers embeddings (local)
- Chroma (local) or Pinecone (hosted) vector store
- Chunked, metadata-tagged HR + IT/Support KB

Why Wilhelm

What differentiates this from a generic LLM chatbot wrapper.

Grounded, not guessed

Every answer traces to a retrieved document. When the knowledge base doesn't cover it, Wilhelm says so — instead of confidently inventing a policy.

Zero provider lock-in

The same codebase runs on OpenAI, Gemini, or a fully offline local model — swappable via one environment variable, not a rewrite.

One answer, five formats

The same grounded answer becomes valid JSON, XML, a downloadable Excel file, or a ready-to-send email — on request, in the same conversation.

Architecture you can see

The live trace panel shows Router → Retrieval → Grounding → Format happening in real time — the pipeline isn't a black box.